INCH-POUND

MIL-DTL-45144F
w/AMENDMENT 1
8 November 2018
SUPERSEDING
MIL-DTL-45144F
7 June 2017

DETAIL SPECIFICATION

FUSE, BLASTING, TIME – M700, AND DUMMY FUSE, BLASTING, TIME - INERT LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and verification methods for the following items: Fuse, Blasting, Time: M700 and Dummy Fuse, Blasting, Time, Inert.

1.2 Classification. The Fuse, Blasting, Time: M700 is a safety type of fuse, suitable for initiating the M7 Non-Electric Blasting Cap. The following are descriptions of the two fuse types and the two constructions:

1.2.1 Type I construction. The fuse consists of a continuous delay train of black powder with three cotton yarns, tightly wrapped and enclosed by ten jute yarns. The fuse is then counter-wound with ten cotton yarns, coated with asphalt material, and covered by a plastic sheath.

1.2.1.1 Type I alternate construction. The fuse consists of a continuous delay train of black powder with one cotton yarn, wrapped with kraft paper, and eight jute yarns. Then the delay train is wrapped in a layer of kraft paper, counterwound with eight jute yarns, and coated with asphalt material. The fuse is then wrapped with eight cottons yarns, coated with asphalt material, and covered by a plastic sheath.

1.2.2 Type II. The Dummy Fuse, Blasting, Time, Inert is of the same configuration as the corresponding Type I except it contains an inert material in place of the black powder.

Comments, suggestions, or questions on this document, should be addressed to the Commander, U.S. Army ARDEC, ATTN: RDAR-EIQ-SE, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Because contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

COMMERCIAL ITEM DESCRIPTION

A-A-208	-	Ink, Marking, Stencil. Opaque, (Porous and Non-porous Surfaces)
A-A-59383	-	Marking Media, Opaque (Porous and Non-Porous Surfaces)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-223	-	Powder, Black
MIL-DTL-15011	-	Pallets, Material Handling, Wood Post Construction, 4-Way Entry
MIL-DTL-2427	-	Box, Ammunition Packing: Wood, Nailed
MIL-C-43438	-	Crimper, Blasting Cap, M-2, Pliers Type. W/Fuse Cutter

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Cards
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil/)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

8830948	-	Cap, Blasting, Non-Electric: M7
8830949	-	Cup
9242364	-	Can, Metal, Packing and Marking, Ammunition for Fuse, Blasting, Time: M700
9242365	-	Box, Wood, Packing and Marking, Ammunition, for Fuse, Blasting, Time: M700
9349653	-	Test Cap, Blasting, Non-electric for M142 F.D.
12972620	-	Igniter, Time, Blasting, Fuse, w/ Shock Tube Capability, M81
13064970	-	Fuse, Blasting, Time: M700 and Dummy Fuse, Blasting, Time: Inert
13073966	-	Box, Wood, Packing and Marking for Dummy Fuse, Blasting, Time: Inert
13074370	-	Box, Fiberboard, Packing and Marking for Dummy Fuse, Blasting, Time: Inert

(These drawings may be requested online usarmy.pica.rdecom-ardec.list.drawing-request-help-desk@mail.mil or from U.S. ARMY ARDEC, RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000.)

US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

19-48-4116	-	Basic Procedures: Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets
19-48-4116/130A	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets: Appendix 130A
19-48-4116/130B	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets: Appendix 130B

(Copies of the above document should be obtained from the contracting activity or as directed by the contracting officer.)

U.S. BUREAU OF ALCOHOL, TOBACCO, AND FIREARMS

Title 27, Code of Federal Regulations

(Copies of this document are available online at www.gpo.gov or from the Superintendent of Documents, U.S. Government Printing Office, North Capitol & "H" Streets, N.W., Washington, DC 20402-0002.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus

(ASTM publications may be ordered online at www.astm.org, or from 100 Barr Harbor Drive PO Box C700, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Design requirements.

3.2.1 Type I fuse construction. The fuse shall be in accordance with all requirements of 13064970 and manufactured such that it complies with the performance requirements of 3.3.

3.2.2 Type II dummy fuse construction. The dummy fuse and its components shall be in accordance with 13064970 and manufactured such that it complies with the performance requirements of 3.3.5 through 3.3.8.

3.3 Performance requirements.

3.3.1 Burning time. The burning time of a 36 inch length of the fuse shall be 120 ±12 seconds at sea level and ambient temperature.

3.3.2 Waterproofness. The M700 shall withstand immersion in water at a minimum pressure of 70 (psig) for not less than 24 hours; afterwards a 36 inch length of fuse shall have the burning time of 120 +12/-50 seconds and initiate an M7 non-electric blasting cap.

3.3.3 Flame intensity. The M700 shall initiate an M7 non-electric blasting cap.

3.3.4 Operating temperatures. The M700 shall exhibit no evidence of damage and shall continue to meet all the performance requirements of 3.3 after being subjected to temperature conditioning at -50˚F, 75˚F, and 125˚F for not less than four (4) hours.

3.3.5 Transit and storage temperatures. The fuse shall exhibit no evidence of damage and shall continue to meet all the performance requirements 3.3 after being subjected to the temperature conditioning at -65ºF and 165ºF for not less than 72 hours.

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3.3.6 M7 attachment integrity. The connection of an M7 Non-Electric blasting cap to the fuse shall withstand a minimum tensile force of 10 pounds applied to the junction of the fuse and blasting cap for not less than five (5) seconds.

3.3.7 M81 attachment integrity. The connection of an M81 Igniter to the fuse shall withstand a minimum tensile force of 15 pounds applied to the junction of the fuse and igniter for not less than five (5) seconds.

3.3.8 Tensile strength. The fuse shall withstand a tensile force not less than 25 pounds for not less than five (5) seconds.

3.4 Ammunition lot numbering. Ammunition lot numbering shall be uniquely assigned to the lot presented for Government acceptance in accordance with MIL-STD-1168.

3.5 Workmanship. All parts and assemblies shall be free of dirt, grease, fungus, and other foreign matter, and defects which could affect their serviceability.

3.5.1 Plastic coating workmanship. The plastic coating shall be smooth, uniform, and free of foreign matter, pin holes, bubbles, breaks, discontinuity, or other defects which could adversely affect the function of the finished fuse.

3.5.2 Marking workmanship. All required markings shall be neat and sharply defined. On the fuse, the markings shall show no lifting, wrinkling, or other visible effect on the plastic's surface. The markings shall be applied on clean, dry surfaces that shall be free of dirt, oil, grease, water, corrosion, or other foreign matter.

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4. VERIFICATION

TABLE I. Requirement/verification cross-reference matrix.

Method of Verification		Classes of Verification						
1- Analysis 2- Demonstration (end item test) 3- Examination 4- Test		A – First Article B – Conformance						
Section 3	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1.1	First article			X	X	X		4.2
3.1.2	Conformance			X	X		X	4.3
3.2.1	Type I fuse construction			X	X	X	X	4.3.3.1, 4.3.3.2
3.2.2	Type II dummy fuse construction			X	X	X	X	4.3.3.7
3.3.1	Burning time				X	X	X	4.5.1
3.3.2	Waterproofness				X	X	X	4.5.2
3.3.3	Flame intensity				X	X	X	4.5.2, 4.5.3
3.3.4	Operating temperatures				X	X	X	4.5.1
3.3.5	Transit and storage temperatures				X	X	X	4.5.4, 4.5.5
3.3.6	M7 attachment integrity				X	X	X	4.5.3
3.3.7	M81 attachment integrity				X	X	X	4.5.3
3.3.8	Tensile strength				X	X	X	4.5.3
3.4	Ammunition lot numbering			X		X	X	4.3.4
3.5	Workmanship			X		X	X	4.5.9
3.5.1	Plastic coating workmanship			X		X	X	4.5.9
3.5.2	Marking workmanship			X		X	X	4.5.9
5.1.1	Metal can (unit pack)			X		X	X	4.2.6.7
5.1.1.1	Salt spray on metal can				X	X		Table II
5.1.1.2	Leakage				X	X	X	4.5.6
5.1.1.3	Moisture content				X	X	X	4.5.8
5.1.2	Wood box (exterior container)			X		X	X	4.3.6.8
5.1.2.1	Marking durability				X	X		4.5.7
5.1.3	Unitization			X		X	X	4.3.3.6
5.2.1	Preservation and packing			X		X	X	4.3.3.9, 4.3.3.8
5.2.2	Unitized load			X		X	X	4.3.3.11

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with either Table II for Type I or Table III for Type II.

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TABLE II. First article tests and inspections (Type I).

Description	Sample size	Requirement	Verification
M700 Time Blasting Fuse (13064970) 1/			
Examination for Defects	10 2/	3.2.1	4.3.3.1, 4.3.3.2
Waterproofness	2/	3.3.2	4.5.2
Tensile/attachment integrities	2/	3.3.6, 3.3.7, 3.3.8	4.5.3
Storage temperatures	2/	3.3.5	4.5.4
Operating temperatures	2/	3.3.1, 3.3.3, 3.3.4	4.5.1
Can, Metal, Packing and Marking, Ammunition, for Fuse, Blasting, Time: M700 (9242364) 1/			
Moisture content	4/	5.1.1.3	4.5.8
Examination for defects	10	5.1.1	4.3.3.3
Salt Spray	5 3/	5.1.1.1	ASTM B117
Leakage	5 3/	5.1.1.2	4.5.6
Box, Packing and Marking, Ammunition, for Fuse, Blasting, Time: M700 (9242365) 1/			
Examination for defects	10	5.1.2	4.3.3.4, 4.3.3.5
Marking durability	5 5/	5.1.2.1	4.5.7
Metal can	PPP-C-96	5.1.1	PPP-C-96
Wood box	MIL-DTL-2427	5.1.2	MIL-DTL-2427
Pallet	MIL-DTL-15011	5.1.3	MIL-DTL-15011

1/ Accept on zero defects; reject on one defect.

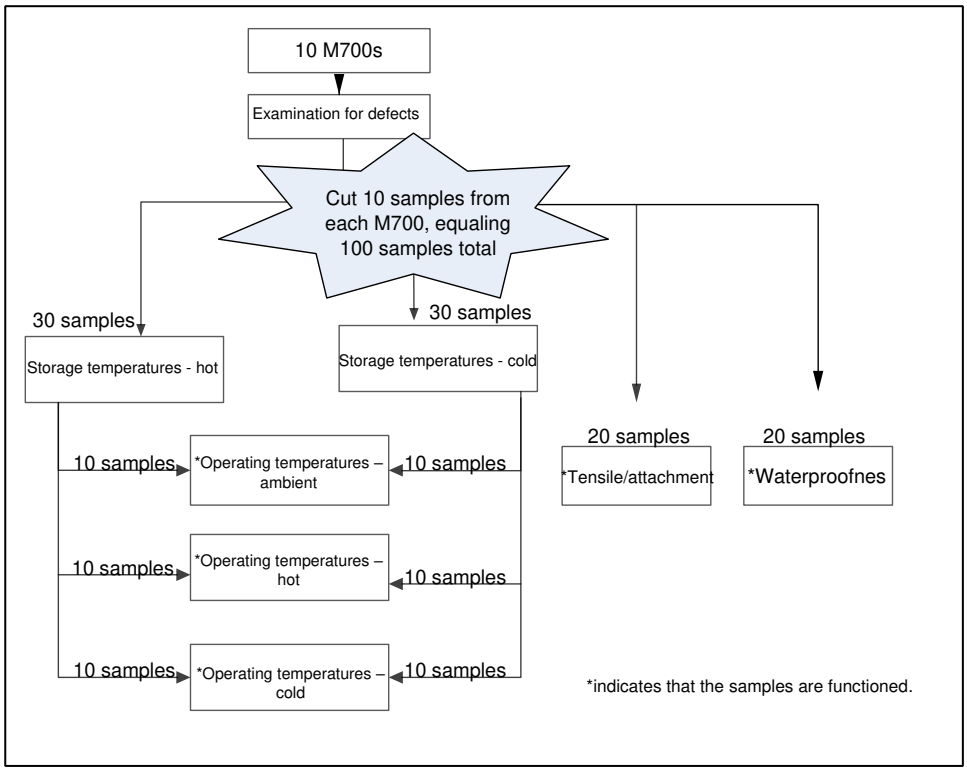
2/ Sequentially test per Figure 4. 100% inspection of all characteristics shall be performed.

3/ Examination for defects shall be performed on the samples prior to testing. Empty cans may be utilized for the Salt Spray test.

4/ Three (3) samples of each type of packaging material shall be randomly selected and subjected to moisture content verification.

5/ Examination for defects shall be performed on the samples prior to testing.

FIGURE 4: FAT sequential testing.



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TABLE III. First article tests and inspections (Type II).

Description	Sample size	Requirement	Verification
Dummy Fuse, Blasting, Time 1/			
Examination for Defects	10	3.2.2	4.3.3.7
Transit/storage temperatures - hot	10	3.3.5	4.5.5
Transit/storage temperatures - cold	10	3.3.5	4.5.5
Tensile/attachment integrities	10	3.3.6, 3.3.7, 3.3.8	4.5.3
Box, Wood and Fiberboard, Packing and Marking, Ammunition, For Dummy Fuse, Blasting, Time: Inert (13073966 & 13074370) 2/	5	5.2.1	4.3.3.8, 4.3.3.9, 4.3.3.10
Pallet	MIL-DTL-15011	5.2.2	MIL-DTL-15011

Notes:

1/ Accept on zero defects; reject on one defect. A total of 10 dummy fuses are required. On each 50ft length of dummy fuse, perform examination for defects and then cut three (3) samples from each. The three samples from each dummy fuse shall be distributed to the Storage Temperatures- Hot, Storage Temperatures- Cold, Marking Durability, and Tensile/Attachment integrity tests.

2/ The verifications shall be performed on five (5) samples. Empty fiberboard boxes may be utilized.

4.2.1 First article rejection. If any assembly, component, or test specimen fails to comply with any of the requirements, the sample shall be rejected.

4.3 Conformance inspection. Conformance samples shall be randomly drawn during production. Conformance inspection for Type I construction shall be performed in accordance with 4.3.3.1 through 4.3.3.6. Conformance inspection for Type II shall be in accordance with 4.3.3.6 through 4.3.3.10.

4.3.1 Conformance rejection. If any assembly, component or test specimen fails to comply any of the requirements, the lot from which the sample was taken shall be rejected.

4.3.2 Lot formation. The lots shall conform to the requirements of MIL-STD-1916. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and revisions, and the same specifications and revisions. In addition inspection lots shall contain parts, components, and subassemblies from not more than of one lot interfix number from one manufacturer.

4.3.3 Classification of characteristics.

a. For the conformance inspection paragraphs 4.3.3.1 to 4.3.3.10, the definitions of critical, major and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.

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b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.

c. For the functional testing, the criteria for switching from normal to reduced sampling shall be a minimum of ten (10) consecutive lots meeting the acceptance criteria for all the inspections and tests. Compliance with the criteria in any one examination/test alone will not result in reduced sampling of that examination/test. Revert to normal sampling upon failure of any one examination/test and continue until the criterion of ten (10) consecutive lots is met again.

d. Alternative conformance acceptance. Alternate conformance procedures may be proposed (see paragraph "Acceptance by contractor-proposed provisions" of MIL-STD-1916).

4.3.3.1 Fuse, Blasting, Time: M700 - examinations.			DRAWING NUMBER 13064970-1 NEXT HIGHER ASSEMBLY 9242364	
CATEGORY	EXAMINATION	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Diameter	III	3.2.1	Gage
102	Length	III	3.2.1	Gage
103	Single band spacing	III	3.2.1	Gage
104	Double band spacing	III	3.2.1	Gage
105	Black powder composition	10	3.2.1	MIL-P-223
Minor				
201	Plastic coating thickness	II	3.2.1	Gage
202	Exterior plastic color	II 1/	3.2.1	Visual
203	Band width	II	3.2.1	Gage
204	Band color	II 1/	3.2.1	Visual
205	Workmanship	II	3.2.1	4.5.9
Notes:	1/ The color shall be visually verified for a general match to the color chip in natural light.			

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4.3.3.2 Fuse, Blasting, Time: M700 - functional.	DRAWING NUMBER 13064970-1
	NEXT HIGHER ASSEMBLY 9242364

CATEGORY	TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Operating temperatures-ambient	10 1/	3.3.4	4.5.1
102	Operating temperatures-hot	10 1/	3.3.4	4.5.1
103	Operating temperatures-cold	10 1/	3.3.4	4.5.1
104	Waterproofness	10 1/	3.3.2	4.5.2
105	Tensile and attachment integrities	10 1/	3.3.6, 3.3.7, 3.3.8	4.5.3
Minor	None defined			

Notes: 1/ 10 M700s total shall be selected for all the testing. Accept on zero defects; reject on one defect.

Perform all examinations per 4.3.3.1 prior to testing.

From each M700, cut five (5) samples for Majors 101; 102; 103; 104; and 105, each no less than 5ft in length. The five samples from each shall be distributed to the Operating temperatures, Waterproofness, and Tensile/attachment integrities.

4.3.3.3 Can, Metal, Packing, Ammunition, for Fuse, Blasting, Time: M700.	DRAWING NUMBER 9242364
	NEXT HIGHER ASSEMBLY 9242365

CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Inside diameter	II	5.1.1	Gage
102	Leakage test	II 1/	5.1.1.2	4.5.6
103	Moisture content	2/	5.1.1.3	4.5.8
104	Can in accordance with PPP-C-96	PPP-C-96	5.1.1	PPP-C-96
105	Can coating complete and correct	II	5.1.1	Visual
Minor				
201	Container undented and undeformed	II	5.1.1	Visual
202	Number of assemblies	II	5.1.1	Visual
203	Marking present, correct, and identifiable	II	5.1.1	Visual
204	Workmanship	II	5.1.1	4.5.9

Notes: 1/ Accept on zero defects; reject on one or more defects.

2/ At the start and after the middle of each production day, three (3) samples of each type of packaging material shall be randomly selected and subjected to moisture content verification. If any samples are found to be in non-conformance, all packaging since the last acceptable set of three (3) tests shall be rejected

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4.3.3.4 Box, Wood, Packing and Marking, Ammunition, for Fuse, Blasting, Time: M700 (prior to closing).			DRAWING NUMBER 9242365 NEXT HIGHER ASSEMBLY 19-48-4116/130B	
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Box in accordance with MIL-DTL-2427	MIL-DTL-2427	5.1.2	MIL-DTL-2427
Minor				
201	Number of metal cans	II	5.1.2	Visual
202	Contents present	II	5.1.2	Visual
203	Workmanship	II	5.1.2	4.5.9

4.3.3.5 Box, Wood, Packing and Marking, Ammunition, for Fuse, Blasting, Time: M700 (sealed).			DRAWING NUMBER 9242365 NEXT HIGHER ASSEMBLY 19-48-4116/130B	
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Contents secure	I	5.1.2	Manual
102	Strapping proper, unbroken and tight	I	5.1.2	Manual/Visual
103	Box undamaged so its contents are not exposed	I	5.1.2	Visual
104	Markings –including 2D bar codes along with wood preservative, heat treatment and POP –present, correct, legible, correct size/height, and in proper location	I	5.1.2	Visual
105	Board, unbroken and unsplit	I	5.1.2	Visual
Minor				
201	Strapping properly engaged	I	5.1.2	Manual/Visual
202	Metallic seal present and properly sealed and positioned	I	5.1.2	Manual/Visual
203	Workmanship	I	5.1.2	4.5.9

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4.3.3.6 Unitized Load, Fuse, Blasting, Time: M700 & Dummy Fuse, Blasting, Time: Inert.			DRAWING NUMBER 19-48-4116/130A 19-48-4116/130B NEXT HIGHER ASSEMBLY 13026660 13073965	
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Pallet in accordance with MIL-DTL-15011	MIL-DTL-15011	5.1.3, 5.2.2	MIL-DTL-15011
102	Pallet load markings -including 2D bar codes along with wood preservative and heat treatment markings - present, correct, legible, correct size/height, and in proper location	100%	5.1.3, 5.2.2	Visual
103	Boxes correctly oriented and positioned on pallet	100%	5.1.3, 5.2.2	Visual
104	Filler assembly made, treated and marked correctly – if required	100%	5.1.3, 5.2.2	Visual
105	Strapping correctly routed, located correctly and material correct	100%	5.1.3, 5.2.2	Visual
106	Unit load tightly strapped and all straps tight	100%	5.1.3, 5.2.2	Visual
Minor				
201	Workmanship	100%	5.1.3, 5.2.2	4.5.9

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4.3.3.7 Dummy Fuse, Blasting, Time: Inert.			DRAWING NUMBER 13064970-2 NEXT HIGHER ASSEMBLY 13074370	
CATEGORY	TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Tensile and attachment integrities	10/6 1/	3.3.8	4.5.3
102	Diameter	II	3.2.2	Gage
103	Length	II	3.2.2	Gage
104	Single band spacing	II	3.2.2	Gage
105	Double band spacing	II	3.2.2	Gage
106	Exterior color	II 2/	3.2.2	Visual
Minors				
201	Band width	II	3.2.2	Gage
202	Band color	II 2/	3.2.2	Visual
203	Workmanship	II	3.2.2	4.5.9

Notes: 1/ Accept on zero defects; reject on one or more defects.
The functional test quantities are depicted as Normal/Reduced. The criteria for switching between Normal to Reduced shall be in accordance with 4.3.3.c. Use the same samples to perform the marking durability, construction, and tensile and attachment integrities tests.

2/ The color shall be visually verified for a general match to the color chip in natural light.

4.3.3.8 Box, Fiberboard, Packing and Marking for Dummy Fuse, Blasting, Time: Inert.			DRAWING NUMBER 13074370 NEXT HIGHER ASSEMBLY 13073966	
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major	None defined			
Minor				
201	Number of dummy fuses	II	5.2.1	Visual
202	Marking present, correct, and identifiable	II	5.2.1	Visual
203	Workmanship	II	5.2.1	4.5.9

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4.3.3.9 Box, Wood, Packing and Marking for Dummy Fuse, Blasting, Time: Inert (prior to closing).			DRAWING NUMBER 13073966 NEXT HIGHER ASSEMBLY 13073965	
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined.			
Major	None defined.			
Minor				
201	Number of fiberboard boxes	I	5.2.1	Visual
202	Marking present, correct, or identifiable	I	5.2.1	Visual
203	Workmanship	I	5.2.1	4.5.9

4.3.3.10 Box, Wood, Packing and Marking for Dummy Fuse, Blasting, Time: Inert (sealed).			DRAWING NUMBER 13073966 NEXT HIGHER ASSEMBLY 13073965	
CATEGORY	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Contents secure	I	5.2.1	Manual
102	Strapping undamaged and secure	I	5.2.1	Manual/Visual
103	Box, undamaged, with its contents not exposed	I	5.2.1	Visual
104	Board, unbroken and unsplit	I	5.2.1	Visual
105	Marking present, correct, and identifiable	I	5.2.1	Visual
Minor				
201	Strapping properly engaged	I	5.2.1	Manual/Visual
202	Metallic seal, properly sealed and positioned	I	5.2.1	Manual/Visual
203	Workmanship	I	5.2.1	4.5.9

4.3.4 Ammunition lot numbering. Visually verify that an ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.4 Test procedures.

4.4.1 Inspection for damage. The entire sample shall be visually inspected for evidence of burning, energetic leakage, being intact and complete, and surface degradation such as separation,

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cracking, or bubbling. The fuse shall be rejected if any of the said defects are found.

4.5 Methods of verification.

4.5.1 Operating temperatures test. The test shall be conducted at sea level. Divide the samples into the three temperature groups per Table II and 4.3.3.2, as applicable. The samples tested at Hot shall be conditioned at 120º +5º/-0˚F. The samples tested at Ambient shall be conditioned at 75º +10º/-0˚F. The samples tested at Cold shall be conditioned at -35º +0˚/-15ºF. Each sample shall be conditioned for a minimum of 4 hours at their respective temperature. Within 10 minutes after the removal of the conditioned samples from the chamber, the sample shall be inspected in accordance with 4.4.1, a 36.0 ±0.50 inch section shall be cut from the approximate middle of the sample, double crimp an M7 blasting cap in accordance with 8830948 or an equivalent test cap in accordance with 9349653 using an M2 type cap crimper in accordance with MIL-C-43438, attach an M81 igniter, and initiate the igniter. Record all burn times. The M700 shall be rejected if the Ambient sample(s) does not meet the burning time requirement or if the Hot or Cold sample(s) does not have a spit of flame exiting the end of the fuse after the igniter is functioned.

4.5.2 Waterproofness test. At the approximate middle of each sample, cut one 42.00 ±0.50 inch section. The sections shall be subjected to the following tests:

a. Submersion at 70 psig. Seal both ends of each section by inserting on and crimping a cup per 8830949 or approved alternate, using a bench type cap crimper which produces either a double cutthroat or an approved alternate crimp. Tie the test assembly so that each end forms a separate loop. Submerge both loops in a suitable vessel of water, leaving approximately four inches of the ends out of the water. The vessel shall be closed and the pressure within the vessel increased to 75 psig and maintained above 70 psig during the test. The test assembly shall remain in the vessel under pressure for the specified time. After being subjected to the required pressure, remove the test assembly, inspect the test assembly in accordance with 4.4.1, and then conclude the remainder of the waterproofness test within 1 hour.

b. Initiation underwater. Cut the two ends off the test assembly so that 36.0 ±0.5 inch samples remain for testing from the middle of the fuse. The cuts shall be clean and square. Onto the end of the fuse, double crimp an M7 blasting cap in accordance with 8830948 or an equivalent test cap in accordance with 9349653 using an M2 type cap crimper in accordance with MIL-C-43438. Immerse the cap and fuse into 2.5 +0.5/-0 inches of 70º±10˚ F water while keeping the opposite end of the fuse out of the water (approximately 2 inches of fuse). For each sample, insert an M81 on the dry end of fuse. Initiate the M81s. The fuse shall be rejected if the sample(s) do not meet the burning time requirement and/or does not detonate the cap.

4.5.3 Tensile and attachment integrities test. From each of the samples, cut a 36.0 ±0.5 inch section so the ends are cleanly and squarely cut. Subject the sections to the following tests:

a. Tensile integrity. Secure the section approximately 12 inches from one end. Apply the specified load for the specified time approximately 24 inches from the secured end. After the tensile integrity test, the fuse shall be inspected in accordance with 4.4.1. After inspection, unsecure the section.

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b. M81 attachment integrity. Insert one end of the section into the M81, secure the M81 and apply the specified load for the specified time. The fuse shall be rejected if it separates from the M81. Following the M81 attachment integrity test, the section of fuse shall be inspected in accordance with 4.4.1. After inspection, remove the M81 from the section.

c. M7 attachment integrity for M700 (Type I only). Onto the untested end of the sample, double crimp an M7 blasting cap in accordance with 8830948 or an equivalent test cap in accordance with 9349653. Secure the cap and apply the specified load for the specified time to the fuse. The M700 shall be rejected if the fuse separates from the cap. Following the M7 attachment integrity test, the fuse shall be inspected in accordance with 4.4.1. After inspection, attach an M81, and initiate the M81 at sea level. The M700 fuse shall be rejected if the cap(s) fails to detonate.

d. Dummy cap attachment integrity for dummy fuse (Type II only). Onto the untested end of the sample, double crimp a cup per 8830949 or approved alternate. Secure the cup and apply the specified load for the specified time to the dummy fuse. The sample shall be rejected if the dummy fuse separates from the cup.

4.5.4 Transit and storage temperatures test (Type I). Divide the samples into the two temperature groups per Table II and 4.3.3.2, as applicable. The samples tested at Hot shall be conditioned at 160º +5º/-0˚F. The samples tested at Cold shall be conditioned at -50º +0˚/-15ºF. The samples shall be conditioned at Hot/Cold for a minimum of 24 hours. Following the conditioning, the samples shall be subjected to operating temperatures per 4.5.1

4.5.5 Transit and storage temperatures test (Type II). The test shall be conducted on the unpackaged dummy blasting fuse. The samples tested at Hot shall be conditioned at 160º +5º/-0˚F. The samples tested at Cold shall be conditioned at -50º +0˚/-15ºF. The samples shall be conditioned at Hot/Cold for a minimum of 24 hours. Following the storage test, the boxes shall be opened and the dummy blasting fuses shall be visually inspected for evidence of surface degradation such as separation, cracking, or bubbling. The dummy fuse(s) shall be rejected if any of the said defects are found.

4.5.6 Leakage test of metal can. The test shall be conducted on the unit packaged (metal can) M700. The metal can shall be placed in a test chamber and a measured volume of air introduced into the chamber to provide the required pressure for the specified time in accordance with 9242364. The metal can(s) and its M700(s) shall be rejected if the required pressure is not achieved or there is a decrease in pressure.

4.5.7 Correctness and durability of markings. Packaging shall be visually inspected for presence and correctness of markings and that markings are clear and not smeared. If using A-A-59383, the markings shall meet the performance requirements of A-A-59383 when using actual parts or test samples made from the same materials of the end item to be marked. If using the alternative marking specification, A-A-208, the packaging shall demonstrate resistance to rubbing as tested in accordance with A-A-208, paragraph 5.4, fifteen minutes after marking the packaging. Packaging with missing, illegible, or incorrect markings or those whose marking(s) do not meeting the durability tests for resistance to rubbing shall be classified as nonconforming and rejected.

4.5.8 Packaging material moisture content test. At the start and after the middle of each

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production day, three (3) samples of each packaging material shall be randomly selected and subjected to moisture content analysis. Ten (10) grams of each sample shall be accurately weighed and placed in a tared weighing dish. Each dish and contents shall be weighed and placed in an oven and dried at 100 ± 2°C, or 212 ± 4°F, for 2 hours. Each dish and contents shall be cooled in a desiccator for a minimum of 15 minutes and weighed. The loss in weight shall be calculated as percent moisture in each sample. If any sample(s) are found to be in non-conformance, all packaging since the last acceptable set of three (3) samples shall be rejected.

4.5.9 Workmanship. Verify by visual inspection. The sample(s) shall be rejected if any nonconformance is found.

5. PACKAGING

5.1 Type I.

5.1.1 Metal can (unit pack). The components, packing, and marking of the metal can shall be in accordance with 9242364 and be securely packaged.

5.1.1.1 Salt spray on metal can. The sealed metal can shall meet the salt spray requirements on 9242364.

5.1.1.2 Leakage. The sealed metal can shall be in accordance with the leakage requirements of 9242364.

5.1.1.3 Moisture content. The paper packing materials shall be in accordance with the moisture content requirements of 9242364.

5.1.2 Wood box (exterior container). The components, packing, and marking of the wooden box shall be in accordance with 9242365 and be securely packaged.

5.1.2.1 Marking durability. The markings shall be in accordance with applicable ink/paint specification or drawing.

5.1.3 Unitization. The unit load shall be in accordance with 19-48-4116 and 19-48-4116/130B, securely strapped, and properly marked.

5.2 Type II.

5.2.1 Preservation and packing. The packing and marking shall be in accordance with 13074370 and 13073966 to include a 2D bar code, except no Performance Orientated Packaging (POP) marking shall appear on the box.

5.2.2 Unitized load. The unit load shall be in accordance with 19-48-4116 and 19-48-4116/130A, securely strapped, and properly marked.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use.

6.1.1 M700 time blasting fuse is a flexible, waterproof cord that burns at a steady rate and is used for igniting military explosives. It contains military characteristics beyond those used in industry.

6.1.2 The dummy time blasting fuse is an inert replica of the M700 and is used for demonstration and classroom training.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and of all reference documents cited in Section 2 and applicable documents from Section 6.

b. Requirement for first article inspection.

c. Requirement for ammunition data cards as specified in MIL-STD-1168.

d. Requirement for acceptance inspection equipment (AIE).

e. Requirement for evidence that the black powder is manufactured within the last 2 years.

f. Type I COAI: 13026660

g. Type II COAI: 13073965

6.3 Definitions.

6.3.1 Bitumen. Bitumen and bituminous materials are referred to as asphalt.

6.3.2 Right hand twist. Yarn has right hand or “S” twist, if when held vertically, the spirals or twists are seen to incline upward in a left hand direction.

6.3.3 Left hand twist. Yarn has left hand or “Z” twist, if when held vertically, the spirals or twists incline upward in a right hand direction.

6.3.4 Delay train. The delay train consists of the cotton center longitudinal yarns and black powder.

6.4 Burn time variation. The burn time of the fuse is dependent on altitude and temperature conditions. A test burn is conducted before fuse usage to calibrate the atmospheric conditions effect on the burn rate.

6.5 Supersession data. This specification is an amendment to MIL-DTL-45144F.

6.6 Revision overview of changes. Changes to this specification can be found in ECP R18B2002.

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6.7 Subject term (key word) listing.
Demolition
Safety fuse

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations

Custodian:
Army - AR
Navy – AS
Air Force - 70

Preparing activity:
Army - AR
(Project 1375-2018-007)

Review activities:
Navy – MC, SH

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